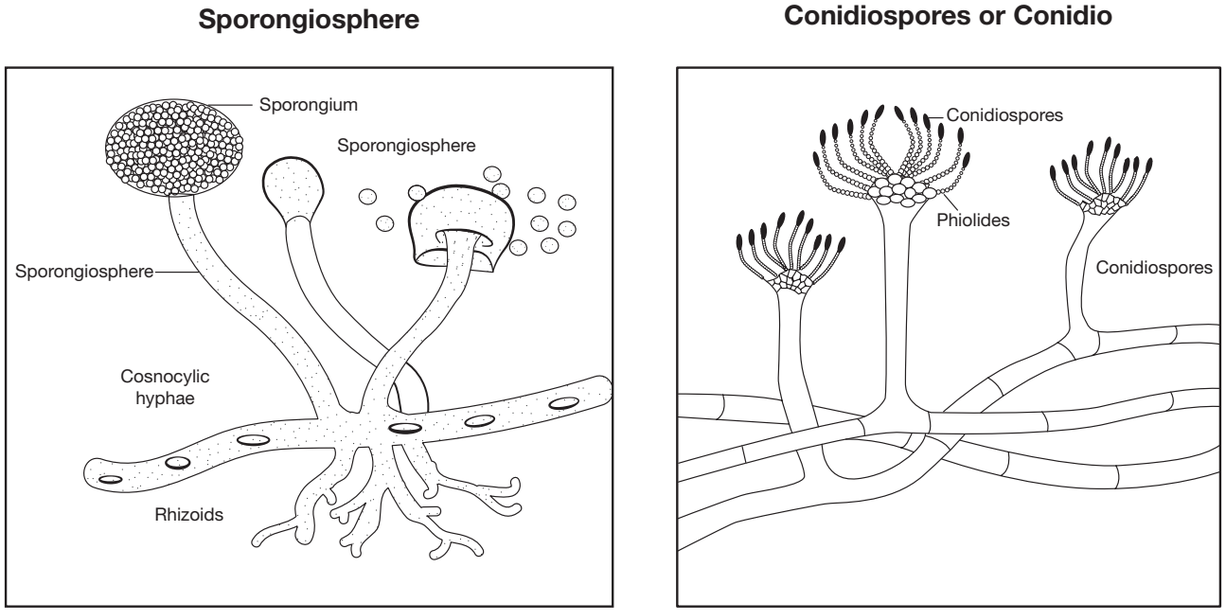


Figure 9.4: Asexual Spores of fungi

known as **sporangiospores** and those that are borne exogenously are called **conidiospores** (Figure 9.4)**.** Based on the arrangement of conidia they are classified as Acropetal, Basipetal and Sympodial.

b. Sexual Reproduction: The process of sexual reproduction typically consists of plasmogamy (cytoplasmic fusion), Karyogamy (union of two nuclei) and meiosis (haploid formation). Anamorphs and Telomorphs are the 2 phases of sexual reproduction

c. Mycelia Sterile: Mycelia sterile are fast growing molds that do not produce spores or conidia. They are medically significant fungi and are difficult to identify

iv. Growth and nutrition

Fungi are ubiquitous in nature and grow readily in the presence of nitrogen and carbohydrates. Medically significant fungi are Mesophilic. The optimum temperature *invitro* for majority of the pathogenic fungi is between 25°C and 37°C. The fungi prefer acidic pH; do not require light for their growth. All fungi are heterotrophs requiring organic nutrients. They absorb their nutrient and do not ingest food. Medically significant fungi are facultative parasites, capable of causing disease or living on dead organic matter.

9.2 Superficial Cutaneous Mycoses

The superficial cutaneous fungal infections involve the outer most layers of skin and its appendages like hair and nails. The causative agents colonize on epidermis or supra - follicular portions of hair and do not penetrate into deeper layers.

The genus ***Malassezia*** is responsible for the superficial infection of the skin. *Malassezia furfur* is lipophilic yeast. It is a commensal of normal skin in the sebaceous glands of warm - blooded vertebrates. It may be pathogenic under certain conditions usually causing skin

TN_GOVT_XII_Micro_Biology_CH09.indd 192 27-02-2019 14:55:14